CUSTOMER 360 / DATA ENGINEERING

One Customer. Multiple Systems. One Trusted View.

Customer360 dataengineering assignmentby LekhanaVaidhya.

Python · Pandas · DuckDB · Parquet · Pytest

Multi-source integration · quality · identity · lineage · publishing



Fragmented Data Creates Fragmented Customer Intelligence.

FRAGMENTED RECORDS

Duplicate customers

CRM and ERP duplicate the same customer.

Inconsistent information

Customer data may be missing or invalid.

Orphan references

Accountsandtransactionsmaylackcustomer links.

Untraceable sources

Source records are difficult to trace.

TRUSTED VIEW

One customer

Identity,relationships,qualitystatus, and provenance remain connected.

A unified view turns disconnected records into usable customer intelligence.

From Raw Data to Trusted Customer Intelligence.

Generate · Syntheticdata generation

Validate · Ingestionand schema/business validation

Resolve · Identity resolution

Transform · Customer 360 transformation

Trace · Lineage, provenance, and monitoring

Publish · Publishing and quality verification

INPUTS
Customers · Accounts · Orders · Interactions

OUTPUTS
Parquet · DuckDB

SEPARATE PATHS
Valid data · quarantined data · uncertain identity matches

Building Trust Before Transformation.

ENTITY KEYS

Customers

Identityanddemographic records

Primary key

Orders

Customertransaction information

Foreign key → Customers

Accounts

Customer-linked account information

Foreign key → Customers

Interactions

Customerengagement records

Foreign key → Customers

VALIDATION GATES

Schema and required-field validation

Email and phone validation

Duplicate business-key detection

Referential integrity and missing/invalid value handling

Valid records → transformation
Invalid records → quarantine with rejection reasons

Matching Identities Without Blindly Merging Customers.

01

Start with source-system identifiers

Preserve system identifiers as evidence.

02

Normalize email

Compare normalized identity evidence.

03

Check consistency

Name and date of birth

Assign deterministic golden customer IDs

Only when evidence is consistent.

Consistent identity evidence
Automatic match

Conflicting or incomplete evidence
Review required

Insufficient evidence
Keep records separate

Missing-email records
Handled separately

Decision boundary: uncertain identities are not automatically merged.

Turning Disconnected Records into a Unified Customer View.

SOURCE RECORDS

Customer Account

Identity record

Linked account information

Order

Transaction information

Interaction

Engagement record

CONVERGE

→

Resolve · combine · validate

GOLDEN CUSTOMER RECORD

One trusted view

Customer, account,order, interaction, aggregated metrics, and latest activity information.

Relationships are validated before publishing. Unresolved references are quarantined.

Every Record Has a Story. Every Run Has Evidence.

1 **Sourcefile**—RunID · sourcefile · sourcerecordidentifier

2 **Ingestion and validation** — Processing stage · validation status · rejection reason

3 **Identity resolution** — Golden customer ID · deterministic evidence

4 **Transformation** — Stage-level logging and metrics

5 **Publishing** — Execution timestamps · output verification

Auditability **Debugging** **Reproducibility**

Designed with Practical Data Engineering Tools.

Tool	Engineering use
Python	Orchestration and processing · workflow control
Pandas	Manipulation and transformation
DuckDB	Analytical database and SQL queries
Parquet	Efficient columnar storage
Pytest	Automated testing
Faker	Synthetic data generation

PUBLISHING

11 datasets

Published to two destinations.

Parquet outputs

DuckDB analytical database

Engineering Results Backed by Evidence.

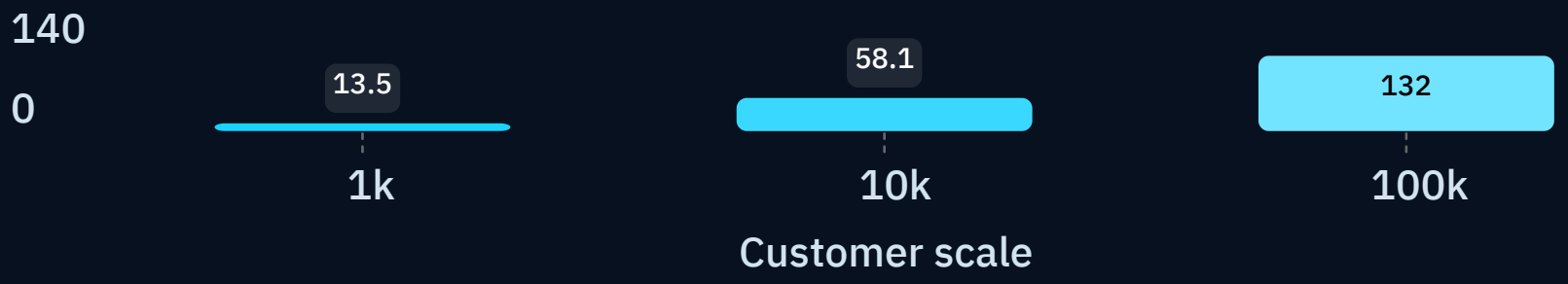
33
automated tests passed

18
data-quality checks passed

11
publisheddatasets reconciled

11
Parquet outputs verified readable

Tests cover validation, identity resolution, transformations, lineage, and stage-level retry behavior. Retry testing supports stage-level behavior; it does not claim complete pipeline recovery.



Benchmark execution time rises with customer scale while remaining measurable across all three benchmark sizes.

A Foundation for Trusted Customer Data.

DELIVERED

Multi-source integration · quality enforcement · deterministic identity resolution

Customer 360 transformation · provenance and lineage · quarantine management

Monitoring, reconciliation, Parquet, and DuckDB publishing

NEXT ENGINEERING IMPROVEMENTS

Atomic publishing across datasets

Checkpoint-based pipeline recovery

Concurrent-run protection

More comprehensive late-arriving data handling

Incremental processing

> “I didn’t just build a data pipeline. I built a traceable data workflow that explains where records come from, how identities are resolved, how data quality is enforced, and how trusted customer information is produced.”

GitHub: https://github.com/lekhanavaidhyarolla/customer-360-data-engineering